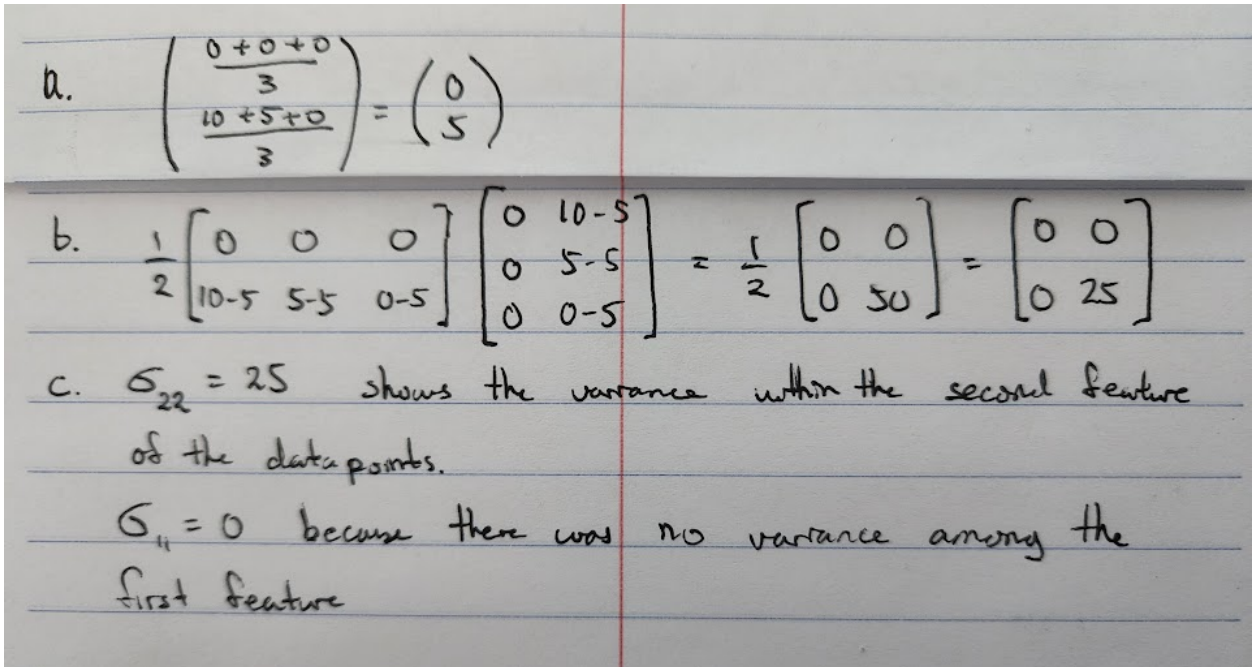
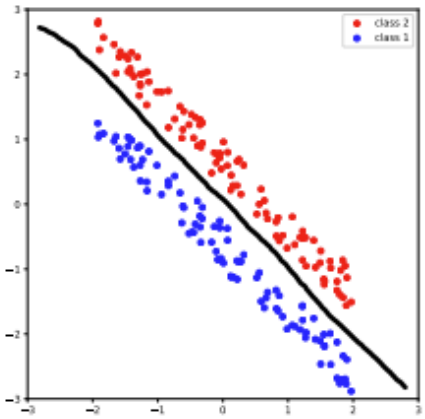
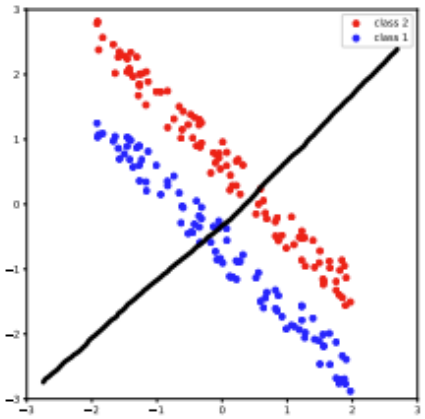
1).



2a).





The PCA method centers and then sets the axis along the most variance in the data. The Linear discriminant chooses a direction that lets data of a class cluster closer together when projected onto the line.

2b).
We should use LDA because it will clearly allow the two classes to become separately grouped in the scenario that we only use 1 linear discriminant.

3). A